SOMAIYA VIDYAVIHAR UNIVERSITY

Somaiya School of Basic and Applied Sciences

# SAGE: Scientific Analysis and Gap-discovery Engine

***Project***

***Submitted in the partial fulfilment for the award of the degree of***

**Master of Science in**

**Data Science**

***By***

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# Certificate

**This is to certify that Ms. Aditi Omprakash Uike with seat no - 44033025001 of program Data Science has satisfactorily completed the project course for the degree of Master of Science in Data Science with the project title "SAGE: Scientific Analysis and Gap-discovery Engine" as prescribed by the School as the partial fulfilment of the degree by Somaiya Vidyavihar University, during the Academic year 2026-27.**

**Signature of Project Guide** **Signature of HOD**

**Name and Sign of Examiner**

**Date:**

**Departmental Seal**

# Declaration by the Student

I certify that

a) The work contained in the Project is original and has been done by myself under the supervision of my guide.
b) The work has not been submitted to any other Institute for any degree or diploma.
c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
d) Whenever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them by citing them in the text of the document and giving their details in the references.

**Date:**

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Signature

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# Acknowledgement

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# SAGE- SCIENTIFIC ANALYSIS & GAP DISCOVERY ENGINE

## Part A: AI Use Case Profile

**Use Case Title:** SAGE – Scientific Analysis & Gap Discovery Engine: An Autonomous Multi-Agent Scientific Research Assistant.

**AI Application Vertical:** Artificial Intelligence, Machine Learning, Data Science and Analytics, Generative AI, Natural Language Processing, Scientific Computing, and Knowledge-Based Systems.

**Collaborating Organization:** Not specified. The proposed system is designed as an independent research-support platform that can be adopted by academic institutions, research laboratories, and scientific organizations.

**Beneficiaries:** Biomedical researchers, clinical laboratory investigators, pharmaceutical research and development teams, life science researchers, academic faculty, postgraduate research scholars, and scientific data analysts.

**Project Context:** A locally hosted, LLM-assisted scientific research platform integrating a React frontend, FastAPI backend, specialized analytical agents, statistical computing, and Retrieval-Augmented Generation (RAG) for scientific literature analysis.

**Primary Demonstration Dataset:** A structured pre-clinical drug-toxicity and cellular-viability assay dataset containing experimental parameters such as dosage, temperature, pH, initial cell density, and cell survival rate.

**Model Runtime:** Google Gemini and/or Ollama-based local Large Language Models (LLMs), depending on the selected configuration.

**Application Overview:** SAGE provides an integrated scientific research workspace that transforms structured experimental datasets into statistical findings, research hypotheses, literature-supported insights, potential research gaps, and experiment recommendations.

## 1. Problem Statement

Scientific research increasingly depends on the analysis of experimental data to understand biological processes, identify relationships between variables, evaluate treatment responses, and support evidence-based conclusions. Research datasets may contain multiple experimental parameters, observations, measurements, and outcomes collected under different conditions.

However, collecting experimental data is only one part of scientific research. Researchers must also perform data cleaning, statistical analysis, pattern identification, hypothesis formulation, and comparison with existing scientific literature. These activities often require knowledge of programming, statistical methods, scientific databases, and research methodologies.

A major challenge is that these activities are generally performed using separate tools and workflows. Researchers may use Python or R for statistical analysis, spreadsheets for organizing observations, scientific databases for literature searches, and document editors for preparing reports. Moving between these tools can increase the time and effort required to complete a research investigation.

Another important challenge is connecting experimental findings with existing scientific knowledge. A statistical relationship observed in a dataset does not automatically establish a scientific explanation. Researchers must examine previously published studies to understand whether similar findings have been reported, whether conflicting evidence exists, and which experimental conditions remain insufficiently explored.

Traditional analytical tools can identify patterns and calculate statistical values, but they do not independently connect these findings with scientific literature and possible research opportunities. Similarly, Large Language Models can assist with scientific interpretation, but their responses may contain unsupported statements, incorrect numerical values, or fabricated references if they are not properly grounded.

SAGE addresses these challenges through an integrated multi-agent framework that combines scientific data analysis, statistical computing, hypothesis generation, literature retrieval, research gap identification, and experiment recommendation.

The system aims to provide a structured workflow in which computational analysis produces verifiable numerical findings, while AI agents interpret these findings and connect them with retrieved scientific evidence.

### 1.1 Core Problem

The central problem addressed by SAGE is the lack of an integrated and systematic approach for transforming structured experimental datasets into statistically supported scientific findings, literature-informed hypotheses, and potential research directions.

### 1.2 Problem Dimensions

- **Fragmented research workflow:** Data preparation, statistical analysis, literature review, and report generation are often performed using separate tools.
- **Technical dependency:** Researchers may require programming and statistical knowledge to perform advanced experimental data analysis.
- **Time-consuming analysis:** Cleaning datasets, identifying patterns, testing hypotheses, and reviewing scientific publications can require considerable time.
- **Hypothesis formulation challenge:** Identifying meaningful and testable hypotheses from experimental observations requires both statistical understanding and domain knowledge.
- **Literature integration gap:** Experimental findings need to be compared with previously published research to understand existing evidence and identify unexplored areas.
- **Research gap identification:** Identifying contradictions, limitations, and insufficiently studied experimental conditions across multiple scientific publications is challenging.
- **Reliability of AI-generated information:** Language models may generate incorrect interpretations or unsupported scientific references without appropriate validation.

### 1.3 Scientific Research Context

Scientific experiments frequently involve multiple independent variables and one or more dependent outcomes. For example, a drug-toxicity experiment may examine how dosage, temperature, pH, and initial cell density influence cellular survival.

Understanding these relationships requires several analytical procedures, including descriptive statistics, correlation analysis, regression, statistical hypothesis testing, and visualization.

Even after identifying statistically meaningful patterns, researchers must determine how the findings relate to existing scientific knowledge. This requires retrieving relevant research papers, examining their methodologies and results, and comparing their conclusions with the observed experimental findings.

### 1.4 Why a Multi-Agent Approach is Relevant

Scientific research consists of multiple interconnected tasks that require different forms of processing. Data validation, statistical analysis, hypothesis formulation, literature retrieval, and report generation should not be treated as a single undifferentiated operation.

SAGE follows a specialized multi-agent architecture in which individual agents are responsible for specific research activities.

For example, the Data Agent prepares the experimental dataset, the Pattern Discovery Agent identifies relationships, and the Statistical Analysis Agent performs appropriate statistical tests.

The Hypothesis Agent uses verified analytical findings to formulate testable hypotheses, while the Literature Agent retrieves relevant scientific publications.

The Gap Discovery Agent compares experimental findings with existing literature, and the Experiment Recommendation Agent suggests possible follow-up investigations. Finally, the Report Agent combines the outputs into a structured scientific report.

This modular approach makes the workflow easier to organize, evaluate, and improve.

### 1.5 Intended Improvement

SAGE aims to reduce the effort involved in moving from experimental data to scientific interpretation and potential research directions.

It is intended to support researchers by providing a connected analytical workflow, improving access to statistical and literature-based insights, and making the research process more organized and traceable.

The system is designed as a scientific decision-support assistant rather than a replacement for researchers, domain experts, or laboratory validation.

## 2. Background and Context

Scientific research in areas such as biomedical science, life sciences, pharmacology, and clinical laboratory investigations generates large amounts of structured experimental data.

These datasets may contain measurements related to drug concentration, cellular response, environmental conditions, biological characteristics, and experimental outcomes. Analyzing such information helps researchers understand patterns, evaluate relationships, and investigate scientific questions.

Traditional scientific analysis involves several stages, including data preprocessing, exploratory data analysis, statistical testing, hypothesis formulation, literature review, and interpretation of results. Although various software tools support these individual activities, integrating them into a continuous research workflow remains challenging.

Recent developments in Artificial Intelligence, Machine Learning, and Large Language Models have created new opportunities for supporting scientific research. Machine Learning techniques can identify complex relationships within experimental datasets, while LLMs can assist with interpreting analytical findings and generating structured scientific explanations.

Retrieval-Augmented Generation (RAG) further enables language models to use information retrieved from external knowledge sources rather than depending entirely on their internal knowledge.

SAGE combines these technologies to create a scientific research-support framework that connects experimental data analysis with scientific literature and research gap discovery.

### 2.1 From Traditional Scientific Analysis to AI-Assisted Research

Traditional scientific analysis generally requires researchers to perform different tasks manually or through separate applications.

For example, a researcher studying drug toxicity may first clean the experimental dataset, calculate descriptive statistics, generate visualizations, perform statistical tests, and formulate hypotheses. The researcher must then search scientific databases to compare the findings with previous studies.

AI-assisted research can help connect these stages by automating selected analytical operations and organizing their outputs into a structured workflow.

SAGE aims to support this process through specialized agents that perform data analysis, statistical evaluation, literature retrieval, and research interpretation.

### 2.2 Relevance of Machine Learning and Statistical Computing

Machine Learning and statistical techniques play an important role in understanding experimental relationships.

SAGE incorporates techniques such as:

- Pearson correlation analysis to examine linear relationships between numerical variables.
- Regression analysis to study relationships between experimental parameters and outcomes.
- Four-Parameter Logistic (4PL) dose-response modelling to analyze suitable sigmoidal responses and estimate IC50 values.
- Gaussian Kernel Density Estimation (KDE) to examine data distributions.
- Random Forest-based feature importance to identify variables that contribute to predictive performance.
- Statistical hypothesis testing, including t-tests, Welch's t-test, Mann–Whitney U test, and ANOVA, where the data and assumptions support their use.

These techniques help identify patterns and provide numerical evidence for further scientific investigation.

### 2.3 Large Language Models and Multi-Agent Systems

Large Language Models can understand natural-language instructions, summarize information, and assist with scientific interpretation.

A multi-agent system divides a complex task into smaller responsibilities. Each agent receives relevant inputs and produces structured outputs that can be used by other agents.

In SAGE, the agents work together to transform numerical findings into understandable scientific interpretations, formulate hypotheses, retrieve relevant literature, and identify potential research opportunities.

The use of specialized agents also makes it easier to track the processing stages and identify where an error may have occurred.

### 2.4 Retrieval-Augmented Generation for Scientific Literature

Scientific literature provides the background needed to interpret experimental findings.

RAG combines information retrieval with language generation. Relevant scientific documents are first retrieved from knowledge sources, and the retrieved information is then provided to the language model as context.

SAGE proposes integrating scientific literature sources such as PubMed, Europe PMC, arXiv, and other accessible scientific repositories.

The retrieved documents will provide information such as publication titles, authors, abstracts, publication dates, DOI identifiers, and source links wherever available.

This approach supports evidence-based literature comparison and helps reduce the risk of unsupported scientific explanations. It does not guarantee that every generated interpretation or citation will be correct, so source verification remains important.

### 2.5 Positioning of SAGE

SAGE is designed as an integrated scientific research-support platform that combines statistical computing, Machine Learning, multi-agent collaboration, LLM-based interpretation, and RAG-based literature retrieval.

Its primary contribution is the integration of these capabilities into a connected workflow that supports the journey from structured experimental data to potential scientific discoveries and research directions.

## 3. Objectives

The objectives of SAGE are organized around scientific data understanding, analytical reliability, hypothesis formulation, literature integration, and research decision support.

### 3.1 Primary Objectives

- Develop an integrated platform that allows researchers to upload structured experimental datasets and provide scientific context for analysis.
- Perform data ingestion, schema identification, data-quality assessment, cleaning, and preprocessing of experimental datasets.
- Analyze experimental data to identify statistical relationships, trends, distributions, and significant patterns.
- Develop AI-driven analytical models for predictive analysis, dose-response modelling, and identification of important experimental variables.
- Perform appropriate statistical tests and generate numerical findings, including statistical values, p-values, confidence intervals, and other relevant measures.
- Generate formal and scientifically testable null and alternative hypotheses based on verified analytical findings.
- Integrate scientific literature retrieval through RAG to compare experimental findings with previously published research.
- Identify potential research gaps by examining differences, contradictions, and insufficiently explored experimental conditions in the available literature.
- Recommend possible follow-up experiments, validation approaches, and research directions.
- Generate structured scientific reports containing analytical results, hypotheses, literature evidence, research gaps, and recommendations.

### 3.2 Secondary Objectives

- Provide interactive visualizations for understanding experimental data and analytical findings.
- Support multiple structured datasets and their individual scientific descriptions.
- Maintain traceability between datasets, analytical results, retrieved publications, and generated hypotheses.

- Support locally hosted LLM inference to reduce dependence on external model services where configured.
- Provide an interactive research question interface for exploring analytical findings.
- Enable report export for documentation and further scientific review.
- Include appropriate error handling and data-quality warnings.
- Support human review of hypotheses, literature interpretations, and experiment recommendations.

### 3.3 Stakeholders and Beneficiaries

| Stakeholder | Potential Use of the Platform |
|---|---|
| Biomedical researchers | Analyze experimental observations and investigate biological relationships. |
| Clinical laboratory investigators | Examine laboratory measurements and identify patterns requiring further investigation. |
| Pharmaceutical R&D teams | Study drug-response behaviour and explore toxicity-related experimental findings. |
| Life science researchers | Analyze biological datasets and formulate testable hypotheses. |
| Academic faculty | Demonstrate scientific data analysis, statistical testing, and AI-assisted research workflows. |
| Postgraduate research scholars | Support literature review, hypothesis formulation, and research gap identification. |
| Data scientists | Apply statistical and Machine Learning methods to scientific datasets. |

## 4. Solution Approach

SAGE follows a structured multi-agent research workflow that begins with experimental dataset acquisition and ends with a scientific report containing analytical findings, literature evidence, research gaps, and possible experimental recommendations.

The framework is organized into five interrelated phases. Each phase performs a specific research activity and passes its outputs to the subsequent stages.

### Phase I – Scientific Data Acquisition and Preparation

Structured experimental datasets are uploaded through the SAGE interface. Researchers can provide a short scientific description or experimental context for each uploaded dataset.

The Data Agent examines the dataset structure, identifies column names and data types, and performs data-quality assessment.

Preprocessing operations include handling missing values, identifying duplicate records, checking invalid or infinite values, converting suitable numerical fields, and identifying zero-variance columns.

The original dataset and information about applied transformations should be preserved to maintain analytical traceability.

### Phase II – AI-Driven Scientific Data Analysis

The prepared dataset is passed to the analytical agents for exploratory and predictive analysis.

The Pattern Discovery Agent identifies relationships, trends, distributions, and potentially important experimental variables using suitable statistical and Machine Learning methods.

The Statistical Analysis Agent performs appropriate inferential tests based on the dataset structure, variable types, sample sizes, and relevant statistical assumptions.

For suitable drug-response datasets, the system can apply 4PL dose-response modelling to estimate IC50 and other model parameters.

The resulting numerical findings are generated through Python-based statistical libraries rather than being estimated by the language model.

### Phase III – Hypothesis Formulation and Scientific Literature Retrieval

The Hypothesis Agent uses the verified analytical findings to formulate formal null and alternative hypotheses.

The Literature Agent retrieves relevant scientific publications from sources such as PubMed, Europe PMC, and other accessible scientific repositories.

Retrieval is based on research questions, experimental variables, observed relationships, and scientific terminology.

The retrieved publications provide the evidence required to compare the experimental findings with existing scientific knowledge.

**Phase IV – Research Gap Discovery and Experiment Recommendation**

The Gap Discovery Agent examines the relationship between the experimental findings and the retrieved scientific literature.

It identifies potential areas where evidence may be limited, conflicting, or insufficiently explored.

For example, an experimental dataset may indicate a relationship between drug dosage and cellular survival under a particular temperature condition, while retrieved studies may have examined the same dosage under different environmental conditions.

Such differences may indicate an area requiring further investigation rather than automatically establishing a new scientific discovery.

The Experiment Recommendation Agent uses these observations to suggest possible follow-up experiments, including relevant variables, controls, experimental conditions, and validation approaches.

All recommendations are intended for review by qualified researchers before laboratory implementation.

**Phase V – Scientific Intelligence Dashboard and Report Generation**

The outputs generated by the specialized agents are integrated into a centralized research dashboard.

The dashboard presents dataset information, statistical findings, visualizations, hypotheses, literature evidence, potential research gaps, and experiment recommendations.

An interactive research assistant can help users explore the results and ask questions related to the current study.

The Report Agent combines the outputs into a structured scientific report that can be exported for documentation, academic discussion, and further research review.

## 5. Conceptual Multi-Agent Architecture

SAGE follows a modular multi-agent architecture in which specialized agents collaborate through a coordinated analytical pipeline.

The architecture separates deterministic numerical computation from generative interpretation. Statistical values are calculated using Python-based libraries, while LLMs assist with interpretation, scientific explanation, literature comparison, and report preparation.




**Fig.5.1:** Conceptual Architecture of the SAGE Research Framework

### 5.1 User Interaction Layer

The user interaction layer is developed using React and Vite with Tailwind CSS.

It provides the research dashboard, dataset upload interface, scientific context input, analytical visualizations, research question interface, and report access.

Researchers can upload structured datasets, inspect their information, and initiate the research workflow.

### 5.2 API and Service Layer

The backend is implemented using FastAPI.

It handles communication between the frontend and the analytical agents through REST API endpoints. Pydantic schemas are used to structure and validate requests and responses.

The API layer manages dataset upload, analysis requests, research questions, report export, and project history operations.

### 5.3 Multi-Agent Processing Layer

The multi-agent layer contains specialized agents responsible for different stages of scientific research.

The Coordinator or orchestration mechanism manages the workflow and ensures that relevant agents receive the required inputs.

Not every agent needs to execute for every request. For example, a simple descriptive analysis may not require literature retrieval or experiment recommendation.

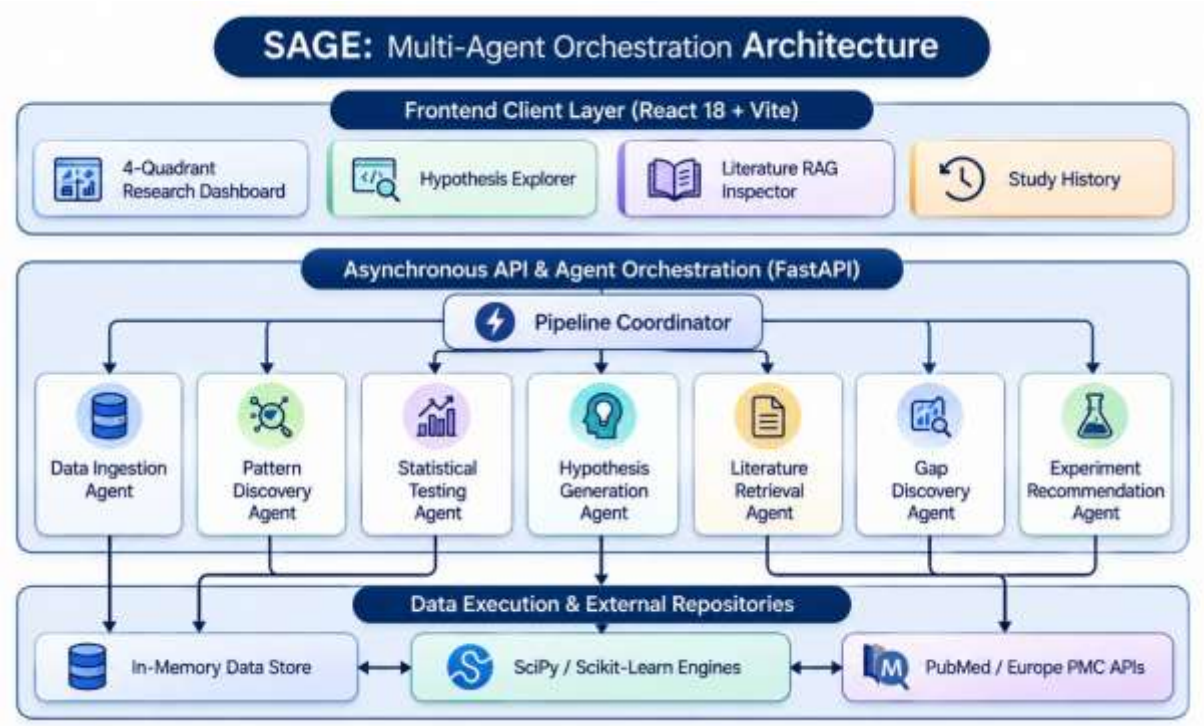



**Fig.5.3.1:** Multi-Agent Orchestration Architecture of the SAGE System

### 5.4 Data and Knowledge Layer

The data layer manages uploaded experimental datasets, processed data, analytical results, and project information.

The literature knowledge layer stores or indexes retrieved scientific documents using a vector database such as FAISS or ChromaDB.

The retrieved literature provides contextual evidence for the hypothesis and research gap discovery stages.

### 5.5 Agent Responsibilities

| Agent / Component | Responsibility |
|---|---|
| Data Agent | Loads structured datasets, identifies schemas, checks data quality, and performs preprocessing. |
| Pattern Discovery Agent | Identifies correlations, distributions, trends, nonlinear relationships, and important variables. |
| Statistical Analysis Agent | Performs statistical tests and calculates numerical measures such as p-values, test statistics, and confidence intervals. |
| Hypothesis Agent | Formulates formal null and alternative hypotheses using verified analytical findings. |
| Literature Agent | Retrieves scientific publications and extracts relevant evidence and citation metadata. |
| Gap Discovery Agent | Compares experimental findings with existing literature to identify potential research gaps and contradictions. |
| Experiment Recommendation Agent | Suggests possible follow-up experiments and validation approaches for expert review. |
| Report Agent | Organizes analytical findings, hypotheses, literature evidence, and recommendations into a scientific report. |

### 5.6 Separation of Responsibilities

The Data Agent focuses on dataset preparation, while the analytical agents perform numerical computations.

The Hypothesis Agent does not independently invent statistical findings. It uses the outputs produced by the statistical and pattern discovery components.

Similarly, the Literature Agent retrieves scientific publications, while the Gap Discovery Agent examines the relationship between retrieved evidence and experimental observations.

This separation improves modularity, traceability, and the ability to evaluate individual components.

## 6. High-Level Workflow

The high-level workflow of SAGE represents the complete process of scientific data analysis, starting from experimental dataset submission and progressing towards statistical evaluation, hypothesis formulation, literature-based validation, research gap discovery, and report generation.

The system follows a modular multi-agent approach in which each specialized agent performs a specific research task. The outputs generated at each stage are passed to the relevant subsequent stages to support systematic scientific analysis.

### 6.1 Dataset Acquisition and Context Submission

The workflow begins when the researcher uploads one or more structured experimental datasets, primarily in CSV format, through the SAGE research dashboard.

- The researcher uploads the required experimental dataset or datasets.
- A short scientific description or experimental context is provided for each dataset.
- The system registers the uploaded files within the research workspace.
- The Data Agent examines the dataset structure, dimensions, column names, and data types.
- Data-quality checks identify missing values, duplicate records, invalid entries, and other potential inconsistencies.
- Appropriate preprocessing operations are performed, and information about the applied transformations is retained.

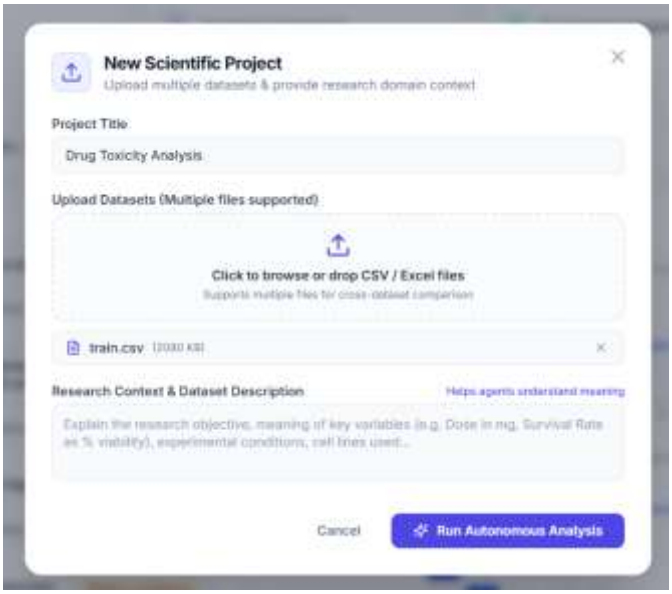



**Fig.6.1.1:** Dataset Upload and Scientific Context Submission Interface

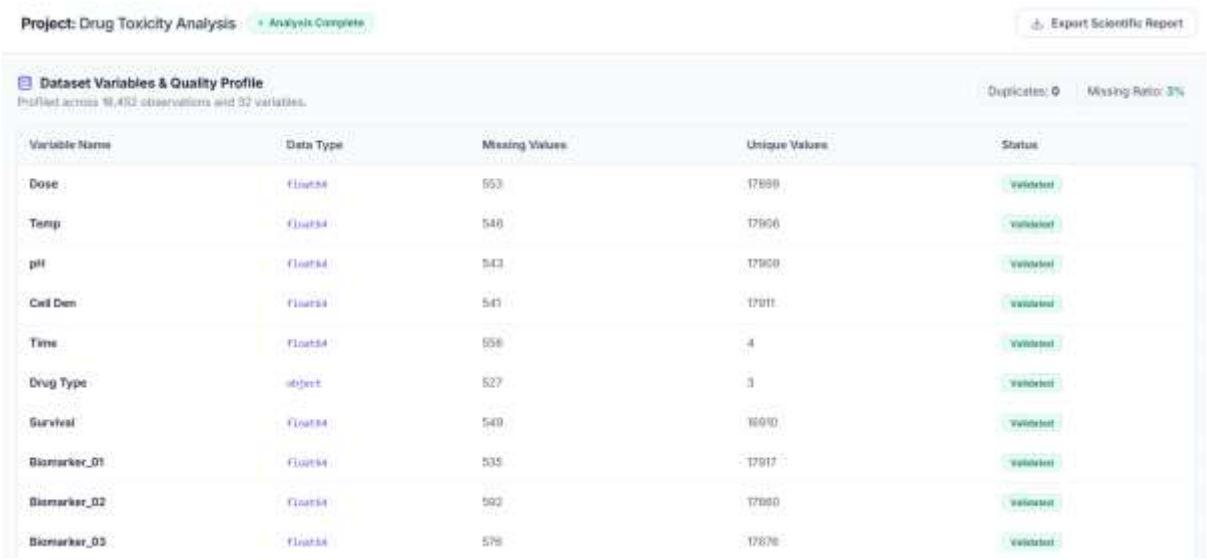



**Fig.6.1.2:** Dataset Preview and Data Quality Assessment Interface

### 6.2 Scientific Analysis and Statistical Evaluation

After dataset validation and preprocessing, the prepared data is passed to the analytical pipeline for scientific exploration and statistical evaluation.

- Exploratory data analysis is performed according to the dataset structure and research context.
- Correlation analysis is used to examine relationships between relevant variables.
- Regression techniques are applied to study trends and relationships between experimental parameters.
- Distribution analysis is performed to understand the spread and behaviour of the data.
- Suitable predictive and analytical techniques are selected according to the research problem.
- Statistical tests such as t-tests, ANOVA, Welch's test, and non-parametric alternatives are applied when appropriate.
- Numerical findings, statistical measures, and relevant visualizations are generated using scientific computing libraries.
- The verified analytical outputs are passed to the subsequent agents for interpretation.

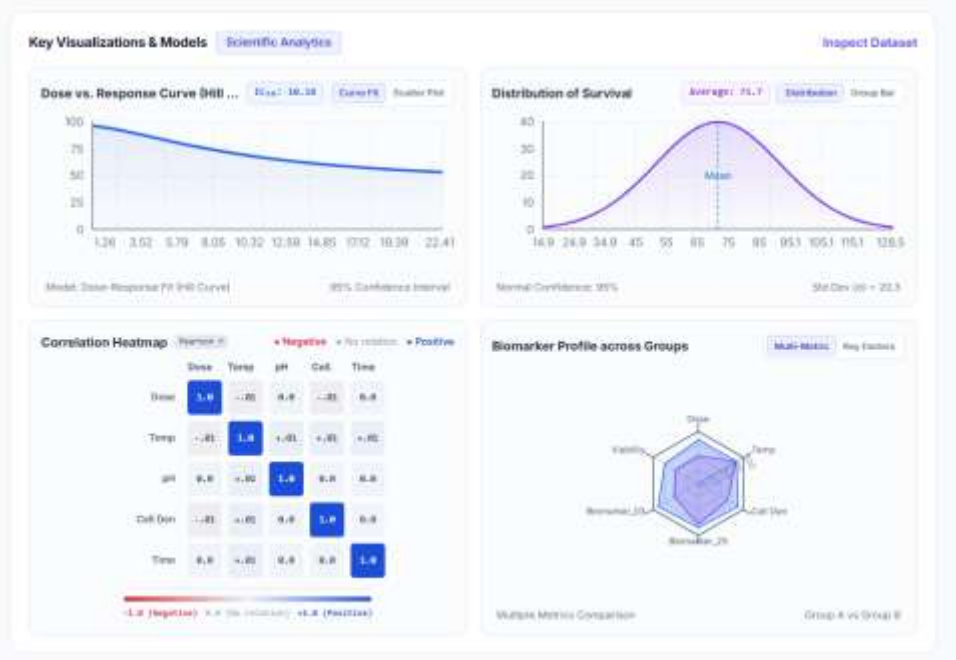



**Fig.6.2.1:** Scientific Data Analysis and Statistical Visualization Dashboard

### 6.3 Hypothesis Generation and Literature Retrieval

The analytical findings are used to formulate research hypotheses and retrieve relevant scientific literature for comparison and contextual understanding.

- The Hypothesis Agent examines the verified statistical findings.
- Formal null hypotheses (H0) and alternative hypotheses (H1) are formulated based on the research question and observed relationships.
- The Literature Agent prepares relevant search queries using the experimental context and analytical findings.
- Scientific publications are retrieved from supported literature repositories.
- Relevant publication details, abstracts, DOI identifiers, and source links are organized.
- Retrieved scientific evidence is compared with the experimental findings to identify supporting, contrasting, or inconclusive observations.

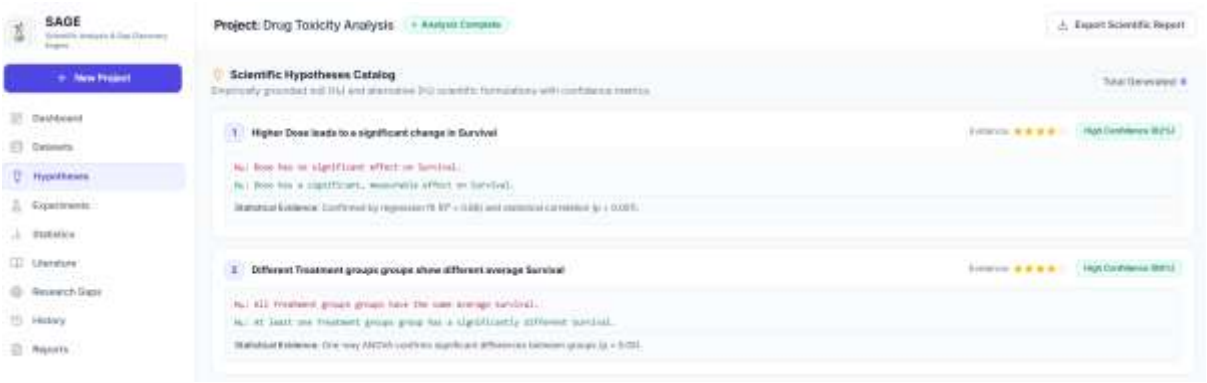



**Fig.6.3.1:** Research Hypothesis Generation Interface

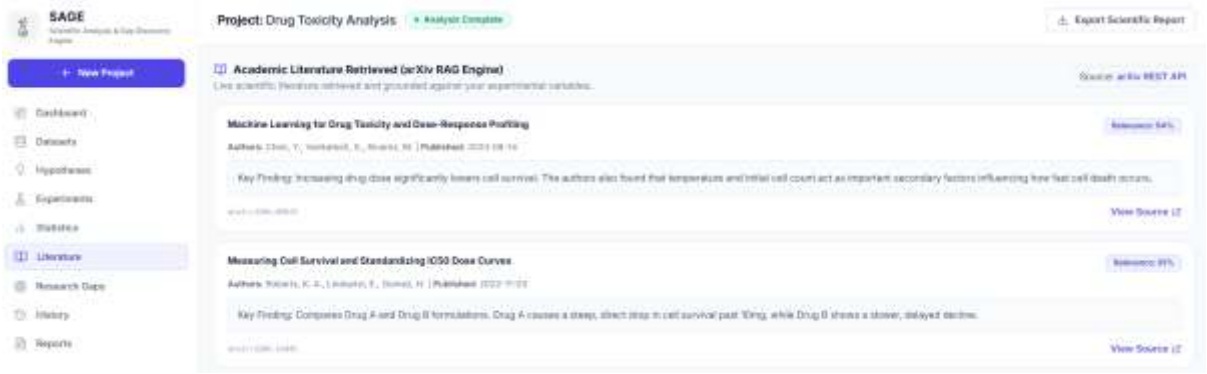



**Fig.6.3.2:** Scientific Literature Retrieval and Evidence Comparison Interface

### 6.4 Research Gap Discovery and Experiment Recommendation

After comparing the experimental findings with the retrieved scientific literature, SAGE proceeds towards identifying possible research gaps and suggesting further investigation.

- The Gap Discovery Agent examines similarities and differences between experimental findings and published research.
- Potential research gaps, conflicting observations, and insufficiently investigated experimental conditions are identified.
- Relevant variables and conditions that may require further investigation are highlighted.
- The Experiment Recommendation Agent prepares possible follow-up experiments and validation suggestions.
- Suggested experimental variables, controls, conditions, and other relevant considerations are presented.
- These recommendations are provided as decision-support suggestions and require review and approval by qualified researchers before practical implementation.

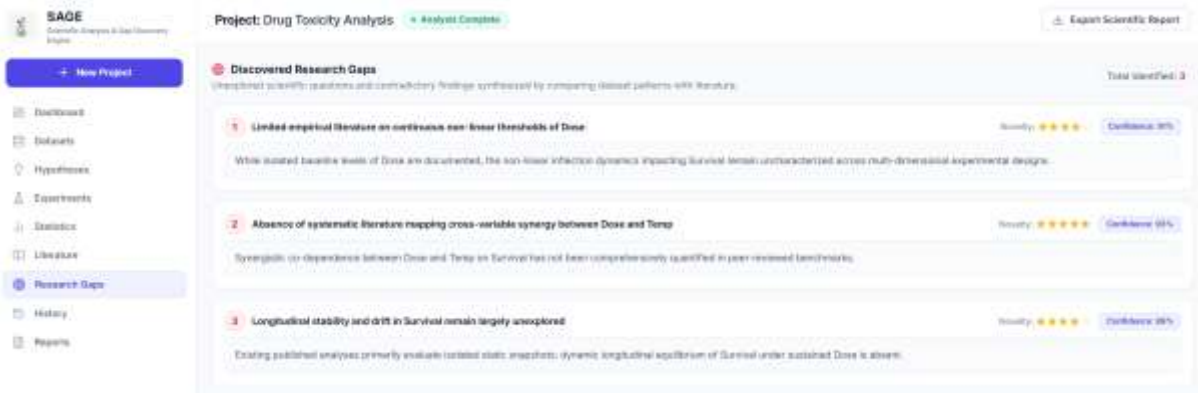



**Fig.6.4.1:** Research Gap Identification and Literature Comparison Interface

### 6.5 Visualization, Reporting, and User Interaction

The final stage integrates the outputs from the different agents and presents them through the SAGE research dashboard.

- Analytical results are organized and displayed through interactive visualizations.
- Graphs and charts represent statistical relationships, data distributions, dose-response behaviour, correlations, and feature importance, where applicable.
- Generated hypotheses, retrieved literature evidence, identified research gaps, and experiment recommendations are presented in an organized manner.
- Researchers can submit additional research-related questions through the interactive research interface.
- The Report Agent combines the available analytical findings and agent outputs into a structured scientific report.
- The generated report is made available for export and further examination by the researcher.

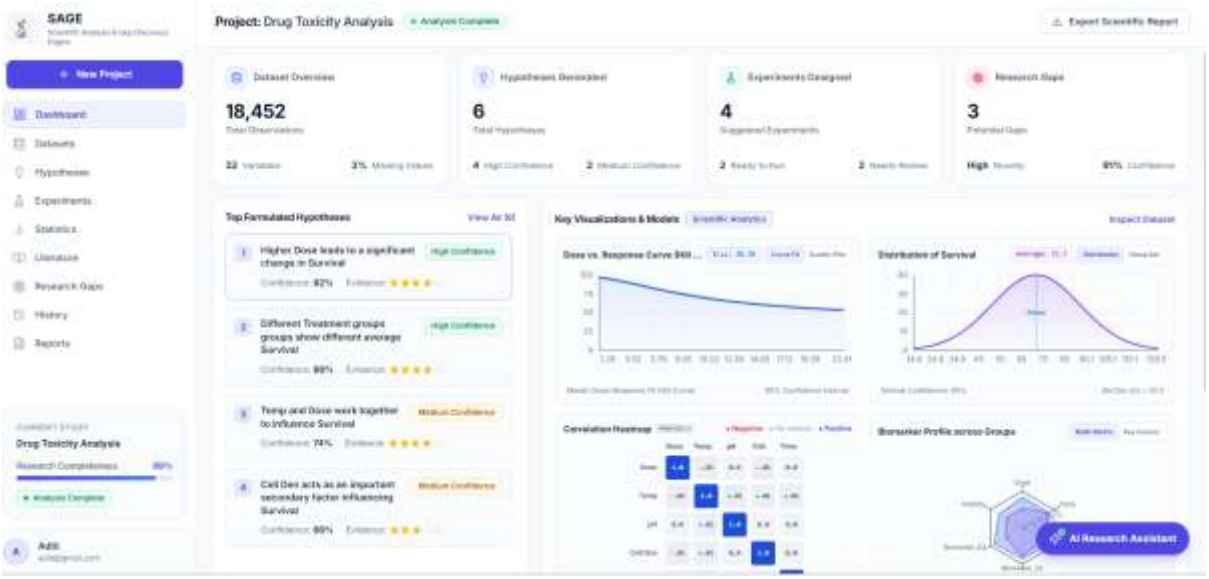



**Fig.6.5.1:** Integrated SAGE Research Dashboard

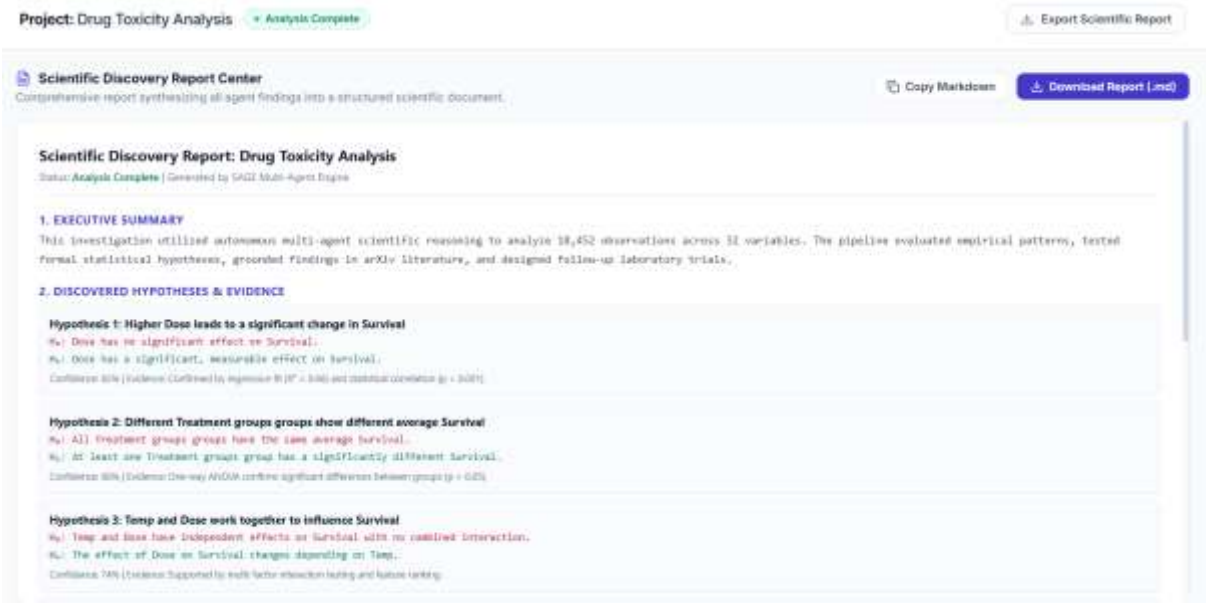



**Fig.6.5.2:** Scientific Research Report Preview and Export Interface

## 7. Expected Outputs

The expected outputs of SAGE are organized into data-management outputs, analytical outputs, scientific interpretation outputs, literature-based outputs, and research decision-support outputs.

### 7.1 Dataset Management Outputs

- Upload and management of structured experimental datasets.
- Support for multiple dataset uploads.
- Scientific description and context associated with each dataset.
- Dataset preview showing the number of observations, columns, and data types.
- Data-quality assessment and preprocessing information.
- Identification of missing values, duplicate records, invalid values, and potential data-quality issues.

### 7.2 Scientific Analysis Outputs

- Descriptive statistical summaries of experimental variables.
- Correlation matrices and relationship analysis.
- Regression and trend analysis.
- Dose-response modelling and IC50 estimation for suitable experimental datasets.
- Distribution analysis using KDE.
- Feature-importance analysis using suitable Machine Learning methods.
- Statistical test results, including relevant test statistics, p-values, and confidence intervals.
- Interactive visualizations of experimental findings.

### 7.3 Hypothesis Generation Outputs

- Formal null hypotheses (H0).
- Formal alternative hypotheses (H1).
- Supporting analytical observations.
- Identification of relevant experimental variables and relationships.
- Testable research questions based on observed findings.

### 7.4 Literature Analysis and Research Gap Outputs

- Retrieved scientific publications from supported repositories.
- Publication metadata, abstracts, DOI identifiers, and source links where available.
- Comparison of experimental findings with published evidence.
- Identification of supporting, conflicting, and inconclusive literature evidence.
- Potential research gaps and insufficiently explored experimental conditions.

### 7.5 Experiment Recommendation Outputs

- Suggested follow-up experiments.
- Relevant experimental variables and conditions.
- Possible control groups and validation approaches.
- Suggested experimental considerations, resources, and timelines where sufficient information is available.
- Recommendations requiring review and approval by qualified researchers.

### 7.6 Scientific Reporting and Decision-Support Outputs

- Centralized research dashboard.
- Interactive research question interface.
- Integrated analytical summaries and visualizations.
- Hypothesis and literature evidence sections.
- Potential research gap summaries.
- Experiment recommendation summaries.
- Exportable scientific research report.

The above outputs describe the intended capabilities of the SAGE framework. Their performance and reliability will need to be evaluated through systematic testing using suitable scientific datasets and verified literature sources.

## 8. Operational Challenges

An autonomous scientific research assistant must handle challenges related to experimental data quality, statistical validity, literature retrieval, AI-generated interpretation, and multi-agent coordination.

### 8.1 Experimental Data Quality

Scientific datasets may contain missing observations, duplicate records, inconsistent units, incorrect data types, and measurement errors.

Such issues can affect statistical calculations and model performance. The system must identify data-quality problems and clearly communicate any preprocessing operations performed.

### 8.2 Statistical Validity

Selecting an inappropriate statistical test can lead to misleading scientific conclusions.

The system must consider sample size, variable types, independence, distributional assumptions, and other relevant statistical requirements before selecting an analytical method.

Small sample sizes or insufficient observations may prevent reliable statistical conclusions.

### 8.3 Model Selection and Performance

Different experimental datasets may require different analytical models.

For example, dose-response modelling may be suitable for sigmoidal biological responses but may not be appropriate for every experimental relationship.

The system must select methods according to the dataset characteristics and avoid producing unsupported predictions when the available observations are insufficient.

### 8.4 Literature Retrieval and Citation Reliability

Scientific literature repositories may contain incomplete metadata, inaccessible publications, duplicate documents, or irrelevant search results.

The system must retrieve relevant publications, preserve source information, and verify available DOI identifiers and publication links.

It should not generate fictional papers or present unverified citations as authentic scientific evidence.

### 8.5 Research Gap Identification

Identifying a genuine research gap requires understanding existing studies, their methodologies, limitations, and experimental conditions.

A difference between the experimental dataset and one retrieved publication does not automatically establish a research gap.

The system must present potential gaps with supporting evidence and acknowledge limitations in the retrieved literature.

### 8.6 Multi-Agent Coordination

Multiple agents may produce incomplete, inconsistent, or conflicting outputs.

The orchestration mechanism must manage dependencies, validate agent responses, handle failures, and maintain traceability throughout the research workflow.

### 8.7 Computational and Infrastructure Constraints

Statistical modelling, Machine Learning, document retrieval, vector search, and LLM inference can require considerable computational resources.

Local model execution may be limited by available memory, processor capabilities, and GPU resources.

The system must manage processing time and resource utilization, particularly when handling large datasets or multiple research projects.

### 8.8 Reliability of AI-Generated Scientific Interpretation

LLMs may generate convincing explanations that are not fully supported by the available evidence.

SAGE addresses this risk by separating numerical computation from generative interpretation. Statistical values must originate from computational libraries, while generated explanations must be grounded in verified analytical outputs and retrieved literature.

These safeguards reduce the risk of unsupported outputs but cannot guarantee that every AI-generated statement is correct.

### 8.9 Summary of Operational Risks

| Risk | Possible Effect | Suggested Control |
|---|---|---|
| Missing or inconsistent data | Incorrect analytical results. | Data-quality checks and preprocessing logs. |
| Inappropriate statistical test | Invalid scientific conclusions. | Assumption checks and method selection based on data characteristics. |
| Model overfitting | Poor predictive performance. | Cross-validation and suitable model evaluation. |
| Irrelevant literature retrieval | Weak evidence comparison. | Semantic retrieval and relevance assessment. |
| Fabricated or incorrect citations | Reduced scientific reliability. | Source and DOI verification. |
| Unsupported research gaps | Misleading research directions. | Evidence-based comparison and human review. |
| Agent failure | Incomplete research workflow. | Error handling, validation, and execution tracing. |
| Excessive computational requirements | Slow processing or system failure. | Resource management and optimized execution. |

## 9. High-Level Workflow – End-to-End Research Process

The complete SAGE workflow can be represented as a connected process in which data analysis, statistical evaluation, hypothesis formulation, literature retrieval, and research gap discovery work together.

### 9.1 Demonstration Use Case

The proposed demonstration focuses on a pre-clinical drug-toxicity and cellular-viability dataset.

The dataset contains experimental variables such as:

- Dosage
- Temperature
- pH
- Initial Cell Density
- Survival Rate

The objective is to examine how experimental conditions relate to cellular survival and identify potential relationships that require further scientific investigation.

### 9.2 Demonstration Workflow

1. The researcher uploads the experimental CSV dataset and provides its scientific context.
2. The Data Agent identifies the available variables and evaluates dataset quality.
3. The dataset undergoes appropriate preprocessing and exploratory analysis.
4. The Pattern Discovery Agent examines relationships between experimental parameters and cellular survival.
5. Suitable regression and dose-response models are applied where supported by the data.
6. The Statistical Analysis Agent performs appropriate statistical tests and calculates numerical findings.
7. The Hypothesis Agent formulates null and alternative hypotheses based on the verified results.
8. The Literature Agent retrieves relevant studies related to drug dosage, cellular viability, and experimental conditions.
9. The Gap Discovery Agent compares the experimental observations with published scientific evidence.

10. Potential research gaps and unexplored experimental conditions are identified.
11. The Experiment Recommendation Agent suggests possible follow-up experiments for expert evaluation.
12. The dashboard presents the analytical findings, hypotheses, literature evidence, and research recommendations.
13. The Report Agent generates a structured scientific report containing the available outputs.

### 9.3 Conditional Agent Execution

SAGE follows a modular workflow in which agents are activated according to the research task.

For example, a request for descriptive statistics may require only the Data Agent and statistical components. A request for research gap discovery may additionally activate the Hypothesis Agent, Literature Agent, and Gap Discovery Agent.

This conditional execution helps avoid unnecessary processing and supports more flexible scientific investigations.

### 9.4 Traceability and Research Validation

Each major stage should maintain a connection with its input data and generated output.

For example, a hypothesis should be linked to the analytical finding that motivated it, while a research gap should be supported by relevant retrieved publications and their source information.

This traceability helps researchers inspect the reasoning behind the generated results and independently verify important findings.

## 10. Modern AI Governance and Paradigms

AI governance is important in SAGE because the platform processes experimental data and generates scientific interpretations that may influence research decisions.

The system should prioritize data protection, statistical reliability, transparency, responsible AI practices, and human oversight.

| Area | Relevance | Recommended Consideration |
|---|---|---|
| Privacy & Data Protection | High | Protect proprietary experimental datasets, research records, uploaded files, and generated reports through secure storage and controlled access. |
| AI Security & Trustworthiness | High | Validate model outputs, handle incomplete or noisy data, monitor agent failures, and prevent unsupported scientific conclusions. |
| Responsible AI | High | Ensure transparency in analytical methods, communicate uncertainty, distinguish correlation from causation, and avoid presenting predictions as established scientific facts. |
| Generative AI / LLMs | High | Use LLMs for scientific interpretation, hypothesis phrasing, literature summarization, and report generation while grounding responses in verified analytical outputs and retrieved evidence. |
| Agentic AI | Medium | Explore coordinated agent workflows for automated data analysis, literature retrieval, research gap identification, and report preparation, with appropriate validation. |
| Human-in-the-Loop Decision Making | High | Enable researchers and domain experts to review hypotheses, literature interpretations, research gaps, and experiment recommendations before accepting or implementing them. |
| Regulatory / Governance Considerations | Medium | Follow applicable institutional research ethics, data protection, scientific integrity, and cybersecurity requirements, particularly when working with clinical or sensitive research data. |

### 10.1 Privacy and Data Protection

Experimental datasets may contain confidential research information, proprietary measurements, or sensitive participant-related information.

The system should implement controlled dataset access, secure storage, appropriate retention policies, and protection of generated reports.

When using an external LLM service such as a cloud-hosted Gemini model, any information transmitted to that service must be handled according to the applicable data-sharing and privacy requirements. Local model execution through Ollama can reduce external data transfer, depending on the deployment configuration.

### 10.2 Transparency and Explainability

SAGE should provide information about the statistical methods, analytical parameters, and retrieved scientific sources used to generate its outputs.

Researchers should be able to inspect the numerical results, visualizations, hypotheses, and supporting literature rather than relying only on generated explanations.

### 10.3 Reliability and Hallucination Control

The system separates mathematical computation from language generation.

Statistical values such as correlation coefficients, p-values, R-squared values, IC50 estimates, and confidence intervals must be calculated using appropriate numerical libraries.

LLMs should interpret these verified values rather than independently calculate or invent them.

Similarly, literature-based statements should be connected to retrieved publications and traceable source information.

These measures help reduce hallucinations but do not eliminate the possibility of errors.

### 10.4 Responsible Scientific Interpretation

A statistical relationship does not automatically establish causation.

SAGE should clearly distinguish descriptive findings, statistical associations, predictive outputs, and scientifically established conclusions.

Potential research gaps and experimental recommendations should be presented as areas for further investigation rather than confirmed discoveries.

### 10.5 Human-in-the-Loop Validation

Researchers and domain experts must retain responsibility for interpreting important scientific findings and approving experimental decisions.

The system should support human review of statistical assumptions, hypotheses, literature evidence, and proposed experimental procedures before these outputs are used in actual research.

## 11. Technology Domains Mapped

SAGE combines multiple technology domains to support scientific data analysis, AI-driven interpretation, literature retrieval, and research decision-making.

| Technology Domain | Application Layer | How it is used in SAGE |
|---|---|---|
| Programming and Software Development | Software | Python-based analytical agents, backend development, and frontend implementation. |
| Application Development and Web Technologies | Application | React, Vite, Tailwind CSS, and FastAPI for developing the research dashboard and API services. |
| Data Science and Analytics | Analytics | Exploratory data analysis, statistical analysis, regression, correlation, and experimental data interpretation. |
| Artificial Intelligence and Machine Learning | AI | Pattern discovery, predictive modelling, dose-response analysis, and identification of important experimental variables. |
| Natural Language Processing | AI | Understanding research questions, scientific terminology, and natural-language instructions. |
| Generative AI and Large Language Models | AI | Scientific interpretation, hypothesis formulation support, literature summarization, and report generation. |
| Multi-Agent Systems | Architecture | Division of scientific research tasks among specialized agents and coordinated workflow execution. |
| Database and Data Management | Data | Storage and management of experimental datasets, project information, analytical outputs, and literature metadata. |
| Information Retrieval and Knowledge-Based Systems | Knowledge | Retrieval of relevant scientific publications and evidence through RAG. |
| Statistical Computing and Scientific Modelling | Analytics | Statistical hypothesis testing, regression, 4PL dose-response modelling, and confidence interval estimation. |
| Data Visualization | Presentation | Interactive charts, correlation heatmaps, distribution plots, dose- |

| | | |
|---|---|---|
| | | response curves, and feature-importance visualizations. |
| Software Testing and System Monitoring | Operations | Agent validation, error handling, API testing, logging, and workflow monitoring. |
| Data Governance and Cybersecurity | Governance | Secure dataset handling, controlled access, source traceability, and responsible use of research information. |

### 11.1 Relationship Between Technology Domains

SAGE depends on the integration of multiple technology areas rather than a single AI model.

Programming and software development provide the foundation for implementing the platform. Data science and statistical computing transform experimental observations into numerical findings, while Machine Learning techniques identify complex relationships and predictive patterns.

Natural Language Processing and LLMs support scientific question interpretation and explanation. Multi-agent orchestration connects the different analytical responsibilities, while RAG provides access to relevant scientific literature.

Finally, visualization and report-generation technologies communicate the findings in a structured format that researchers can inspect and evaluate.

### 11.2 Educational Relevance

SAGE provides a practical example of how programming, statistical computing, Machine Learning, Generative AI, information retrieval, and software engineering can be integrated into a scientific research application.

The project demonstrates the role of AI in supporting scientific investigation while highlighting the importance of statistical validation, evidence-based interpretation, transparency, and human oversight.

It provides an interdisciplinary learning opportunity connecting computer science, data science, artificial intelligence, and scientific research methodologies.

## 12. Evaluation Methodology

The proposed framework will be tested to check whether it works correctly, produces reliable analytical results, retrieves useful scientific literature, and is easy to use. The evaluation will cover both the performance of individual agents and the working of the complete multi-agent system.

Since the framework uses both statistical methods and AI-based interpretation, the evaluation will check them separately. Numerical results will be verified using established statistical methods, while AI-generated explanations and suggestions will be reviewed for their accuracy and relevance.

### 12.1 Functional Testing

Functional testing will be carried out to check whether all the features of the framework work as expected.

The following features will be tested:

- Uploading and handling multiple CSV datasets.
- Adding scientific context and linking it to the correct dataset.
- Examining datasets and identifying missing, duplicate, and incorrect values.
- Applying suitable statistical and analytical methods.
- Generating null and alternative hypotheses.
- Searching for relevant scientific research papers.
- Identifying possible research gaps.
- Suggesting experiments for further research.
- Displaying analytical results on the research dashboard.
- Generating and exporting scientific reports.

Each feature will be tested using suitable datasets and different test cases. The system should also display clear error messages and warnings when users upload invalid files, unsupported formats, or datasets that do not contain enough information for analysis.

### 12.2 Data Quality and Preprocessing Evaluation

The Data Agent will be tested to check how correctly it identifies data-related problems and prepares datasets for further analysis.

The evaluation will include:

- Correctly identifying the number of rows, columns, column names, and data types.

- Detecting missing values and duplicate records.
- Identifying incorrect, infinite, or inconsistent values.
- Correctly handling numerical and categorical columns.
- Identifying columns that contain only one repeated value.
- Applying suitable data cleaning and preprocessing methods.
- Keeping a record of the original data and the changes made during preprocessing.

The original dataset will be compared with the processed dataset to check whether the cleaning operations have been performed correctly. This will help ensure that no unwanted changes are made to the data.

### 12.3 Statistical and Analytical Evaluation

The statistical analysis component will be tested by comparing its results with manually calculated values or results obtained using trusted statistical software.

The evaluation will cover the following methods:

- **Correlation Analysis:** Checking whether Pearson or Spearman correlation values are calculated correctly for suitable datasets.
- **Regression Analysis:** Checking regression coefficients, R-squared values, and how well the regression model fits the data.
- **Statistical Testing:** Comparing test statistics, degrees of freedom, p-values, and confidence intervals obtained from t-tests, ANOVA, and non-parametric tests.
- **Dose-Response Analysis:** Checking the fitting of the four-parameter logistic (4PL) model and the calculation of IC50 values when the dataset is suitable for this analysis.
- **Distribution Analysis:** Checking whether the distribution graphs and density plots correctly represent the given data.
- **Feature Importance:** Checking whether the selected method correctly identifies important features and provides understandable results.

The system will also be checked to ensure that each statistical method is used only when its required conditions are suitable. For example, a statistical test should not be applied without checking whether the dataset meets the basic requirements of that test.

### 12.4 Hypothesis Generation Evaluation

The Hypothesis Agent will be evaluated based on how relevant, clear, and scientifically meaningful its generated hypotheses are.

The evaluation will check:

- Whether the hypotheses are based on actual findings from the data analysis.
- Whether the null hypothesis (H0) and alternative hypothesis (H1) are clearly explained.
- Whether the hypotheses can be tested through further experiments or statistical analysis.
- Whether the hypotheses match the research question and experimental context.
- Whether the explanations agree with the statistical results.

The generated hypotheses will also be reviewed by researchers or subject experts to check whether they are scientifically relevant and suitable for further investigation.

### 12.5 Literature Retrieval and Research Gap Evaluation

The Literature Agent and Gap Discovery Agent will be tested to check whether they retrieve useful scientific information and identify possible research gaps correctly.

The evaluation will consider:

- Whether the retrieved research papers are relevant to the research question.
- Whether publication details such as paper titles, author names, abstracts, DOI numbers, and source links are correct.
- Whether the information taken from the retrieved publications is represented accurately.
- Whether the system can identify research papers that support or disagree with the observed findings.
- Whether the suggested research gaps are relevant to the topic.
- Whether the identified gaps are supported by the retrieved scientific literature.

Where possible, the retrieved papers will be compared with a manually prepared list of relevant research publications. Researchers will also review the suggested gaps to check whether they are reasonable and supported by existing evidence.

### 12.6 Experiment Recommendation Evaluation

The Experiment Recommendation Agent will be evaluated based on whether its suggested experiments are relevant, understandable, and practically useful.

The evaluation will check:

- Whether the suggested experiments are related to the identified research gaps.
- Whether the selected variables and experimental conditions are suitable.

- Whether important controls and validation steps are included.
- Whether the recommendations are supported by available scientific evidence.
- Whether the suggestions are clear and useful for researchers.

The suggested experiments will be reviewed by qualified researchers before being carried out in a laboratory.

### 12.7 System Performance and Usability Evaluation

The complete application will also be tested to check its speed, stability, reliability, and ease of use.

The following factors will be considered:

- Time taken to upload and validate datasets.
- Time required to complete the full analysis process.
- Response time of the research question interface.
- Ability to handle different datasets and analysis requests without errors.
- Handling of invalid, incomplete, or unsupported inputs.
- Ease of moving between different dashboard sections.
- Clarity of graphs, analytical results, and generated reports.

Performance will be measured using different datasets and testing conditions. The actual execution times and performance results will be recorded after the system has been implemented and tested.

### 12.8 Overall Evaluation Criteria

The following table summarizes the main areas that will be considered while evaluating the proposed framework.

| **Evaluation Area** | **Evaluation Criteria** |
|---|---|
| Functional Testing | Checking whether all required features work correctly. |
| Data Quality | Correct identification and handling of data-related problems. |
| Statistical Analysis | Comparing results with independently calculated values. |

| | |
|---|---|
| Hypothesis Generation | Checking the scientific relevance and testability of hypotheses. |
| Literature Retrieval | Checking the relevance and correctness of retrieved publications. |
| Research Gap Discovery | Checking whether possible research gaps are supported by scientific evidence. |
| Experiment Recommendation | Checking whether suggested experiments are relevant and practical. |
| System Performance | Measuring response time, stability, and reliability. |
| Usability | Checking how easily users can navigate and understand the system. |

The evaluation results will help identify areas where the framework needs improvement. They will also help determine how effectively the system supports different stages of scientific research. No accuracy or performance value will be considered confirmed until it has been measured through actual testing.

## 13. Limitations

Although the proposed framework aims to make scientific data analysis and research easier, it has certain limitations that need to be considered.

### 1. Dependence on Structured Datasets

The current framework mainly works with structured tabular data, especially CSV files. It does not directly analyse raw genomic sequences, protein structures, medical images, or raw mass-spectrometry signals. These data types would first need to be converted into suitable formats before they can be analysed by the system.

### 2. Dependence on Data Quality

The quality of the analysis depends on the quality and completeness of the uploaded dataset. Missing values, measurement errors, small sample sizes, and differences in experimental conditions may affect the reliability of the results.

### 3. Limitations of Statistical Methods

Every statistical method has its own requirements and conditions. Choosing an unsuitable statistical test, using a very small dataset, or having too many unusual values may affect the results. Therefore, statistical methods must be selected carefully, and their results should be interpreted according to their assumptions.

### 4. Limitations of Language Models

Large Language Models may sometimes provide incorrect explanations, misleading interpretations, or scientific information that is not supported by evidence. The framework uses scientific literature and verified numerical results to reduce such errors. However, it cannot completely guarantee that AI-generated information will always be correct.

### 5. Literature Availability and Retrieval Limitations

The quality of literature-based analysis depends on the scientific publications available through the supported research databases and repositories. Some research papers may have restricted access, incomplete publication details, or limited information available for retrieval.

### 6. Research Gap Validation

The research gaps identified by the system are possible areas for further study. However, a difference between the system's findings and previously published research does not

automatically mean that a genuine research gap exists. Researchers must examine the existing literature and validate these suggestions before accepting them.

### 7. Dependence on Computing Resources

Running statistical methods, searching scientific literature, and using language models may require sufficient memory and processing power. The system's speed and performance may vary depending on the computer hardware, dataset size, and language model being used.

### 8. Need for Human Supervision

The framework is designed to assist researchers, not replace them. Generated hypotheses, literature interpretations, and experiment suggestions must be checked by experts before being used for scientific conclusions or actual laboratory work.

### 9. Privacy and Data Security Considerations

The system is designed to support local deployment, which can help researchers handle their datasets within their own environment. However, when an external language-model API or cloud-based service is used, relevant prompts or data may be sent outside the local system. Therefore, data privacy depends on the selected model, its configuration, and the security measures being followed.

## 14. Future Scope

The proposed framework can be improved in the future by adding new features and supporting more types of scientific research. Some possible areas of future development are discussed below.

### 1. Support for Different Types of Scientific Data

Future versions can support additional data formats such as genomic sequences, protein structures, medical images, microscopy images, and mass-spectrometry data. This would allow researchers to work with different types of scientific information through a single platform.

### 2. Development of a Scientific Knowledge Graph

A scientific knowledge graph can be added to show connections between research papers, authors, experimental variables, scientific findings, hypotheses, and research gaps. This would help researchers understand relationships between different studies and find useful connections in existing scientific knowledge.

### 3. Comparison of Multiple Studies and Datasets

The framework can be extended to compare multiple research studies and experimental datasets. This would help identify similar patterns, differences in experimental results, and areas that need further investigation. Such comparisons would be performed only when the datasets have compatible variables, measurement units, and experimental conditions.

### 4. Advanced Experiment Optimization

Methods such as Bayesian Optimization and Active Learning can be introduced to suggest better experimental conditions. These techniques may help researchers select suitable combinations of variables and reduce unnecessary experiments while considering practical laboratory requirements.

### 5. Integration with Laboratory Automation

In the future, the system may be connected with laboratory automation platforms and robotic equipment, such as Opentrons. This could help transfer experiment protocols approved by researchers into automated laboratory workflows.

### 6. Improved Explainable AI

Explainable AI techniques can be added to make the system's results easier to understand. These techniques can explain why certain features are important, how variables are related, and

how the system arrives at its predictions or recommendations. This would help researchers understand the reasoning behind the generated results.

### 7. Improved Scientific Literature Coverage

More scientific databases, research repositories, and publication APIs can be added to improve the availability of research papers. Better citation checking, document retrieval, and source tracking can also be introduced to make literature-based analysis more reliable.

### 8. Collaboration and Research Management Features

Future versions may include researcher accounts, shared workspaces, project sharing, version tracking, and research history. These features would allow students, researchers, and scientific institutions to work together and manage their research projects more easily.

### 9. Better Support for Local Language Models

The framework can be improved by supporting more efficient language models that can run locally on a computer or institutional server. This may reduce dependence on external APIs, provide better control over research data, and improve privacy when working with confidential experimental information.

## 15. Conclusion

The proposed project, SAGE (Scientific Analysis & Gap Discovery Engine), is an autonomous multi-agent framework designed to support different stages of scientific research. It combines data analysis, statistical methods, hypothesis generation, scientific literature search, research gap identification, and experiment recommendations in a single system.

The framework uses different specialized agents, where each agent is responsible for a particular research task. These agents work together to analyse experimental datasets, understand the results, connect the findings with existing scientific literature, and suggest possible directions for further research.

One of the main features of SAGE is that it separates numerical calculations from AI-generated explanations. Statistical values and analytical results are calculated using established Python libraries, while Large Language Models help explain the findings, summarize scientific information, and generate research-related suggestions. This approach aims to make the results more reliable and easier to understand.

The framework aims to reduce the manual effort required to analyse experimental data and connect research findings with existing scientific knowledge. Its interactive dashboard and report-generation features are intended to provide researchers with a clear and organized way to examine their results and explore possible research directions.

Overall, SAGE is proposed as an AI-based research support system that brings together data analytics, multi-agent technology, and scientific literature analysis. Although its findings and recommendations will still require human review and scientific supervision, the framework provides a foundation for making the research process more organized, efficient, and easier to explore.

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**Turnitin Report**

# Aditi Uike

## SAGE Documentation

Project Synopsis

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